

Arterial Aneurysms (Infrarenal & Peripheral)

The Big Picture

An aneurysm = artery dilated to $\geq 1.5\times$ normal, involving **all 3 wall layers**. Commonest in the **infrarenal aorta**. Key exam themes: **true vs false**, **shape**, **size thresholds for rupture/repair**, and the named peripheral sites with their tell-tale percentages.

Definitions to Nail

- **Aneurysm** — $\geq 1.5\times$ normal, all 3 layers.
- **AAA** — infrarenal aorta **>3 cm** (or 50% bigger than proximal).
- **Ectasia** — enlarged but <50% over normal (intermediate).
- **Arteriomegaly** — diffuse enlargement of many segments >50%.
- **Aneurysmosis** — multiple aneurysms, several locations.
- **True** — all 3 layers. **False/pseudo** — wall is only connective tissue/contained haematoma, with a **neck**; usually iatrogenic (arterial access) or anastomotic breakdown.

Shape (matters for rupture)

- **Fusiform** — uniform circumferential dilation (most degenerative/atherosclerotic).
- **Saccular** — focal, eccentric outpouching from ulcer/trauma/infection; **higher rupture risk** → **repair at smaller size**. Renal & cerebral aneurysms are saccular/eccentric.

Causes

Degenerative (age), inflammatory, dissection, trauma, infection, congenital, **smoking**.

Abdominal Aortic Aneurysm (AAA)

- **Where** — 30% infrarenal; **9× more common than thoracic**; incidence +6% per decade after 65.
- **Smoking = single biggest modifiable risk** (dose-dependent: 12× at >1 pack/day × 35 yrs).
- **Rupture risk** — stratified by diameter; jumps at **>5 cm**; saccular worse.
- **Presentation** — usually **asymptomatic**; if symptomatic, new diffuse abdominal/back pain ("different from usual"), pulsatile abdominal sensation/mass in thin patients. History of tobacco, HTN, atherosclerosis.
- **Imaging** — US (screen), **CTA (excellent)**, MRA; contrast angiography invasive/avoided.
- **Management** — no drug reliably slows growth; β -blockers (BP), antiplatelets (thrombosis). Repair: **open vs EVAR (endovascular)**.

Peripheral & Visceral Sites (memorise the numbers)

- **Iliac** — common iliac commonest (with AAA); **internal iliac** ruptures in 40% / 31% death; bilateral common iliac in 50%; external iliac rarely.
- **Femoral** — true aneurysms rare; complications if **>3.5 cm** or thrombus; **88% have a synchronous aneurysm elsewhere**. **Infected pseudoaneurysm from IV drug abuse = most dangerous** → open surgery. Small <2.5 cm observe; pseudo can be treated with **US-guided thrombin injection / compression**.
- **Popliteal (PAA)** — **most common peripheral aneurysm (70%)**; almost exclusively men 65–80; **bilateral 50%**; **30–50% have AAA**. Presents with **limb ischaemia (thrombosis/embolism)**, rarely rupture. Prominent popliteal pulse + gangrene in non-diabetic man = think PAA. **Repair if symptomatic or >2 cm**.

- **Visceral — splenic = commonest visceral (~60%);** women, portal hypertension; **pregnancy = rupture risk;** hilar lesions need splenectomy.
- **Renal — 0.1%;** saccular + **fibromuscular dysplasia;** rupture in **3rd-trimester pregnancy (maternal 70% / fetal 100% mortality);** repair if >2–3 cm, childbearing women, refractory HTN.
- **Cerebral/UE —** intracranial common; carotid rare; subclavian from thoracic outlet repetitive trauma; brachial pseudoaneurysm iatrogenic.

Walk-Away Points

- AAA >3 cm; **rupture risk climbs >5 cm;** smoking = top modifiable risk; CTA + EVAR/open.
- **Popliteal = 70% of peripheral,** male, bilateral 50%, presents as ischaemia.
- Femoral: **88% have another aneurysm;** infected IVDU pseudoaneurysm most dangerous.
- Splenic = commonest visceral; renal aneurysm + pregnancy = catastrophic rupture.
- Saccular ruptures earlier → repair at smaller diameter.

Arterial Aneurysms — Revision Table	
Term	Definition
Aneurysm	≥1.5× normal diameter, all 3 layers
AAA	Infrarenal aorta >3 cm
Ectasia	Enlarged but <50% over normal
Arteriomegaly	Diffuse multi-segment >50%
True	All 3 layers
False/pseudo	CT/haematoma wall + neck; iatrogenic/anastomotic
Shape	Features
Fusiform	Uniform, circumferential, atherosclerotic
Saccular	Focal eccentric; higher rupture → repair smaller ; renal/cerebral
AAA	Detail
Frequency	30% infrarenal; 9× > thoracic; +6%/decade after 65
Top risk	Smoking (dose-dependent)
Rupture	↑↑ when >5 cm ; saccular worse
Presentation	Asymptomatic; pulsatile mass (thin)
Imaging	US screen, CTA excellent , MRA
Repair	Open vs EVAR
Peripheral site	Key numbers / facts
Popliteal (PAA)	70% of peripheral , men 65–80, bilateral 50%, 30–50% +AAA; presents as ischaemia; repair if Sx or >2cm
Iliac	Common iliac commonest; internal iliac 40% rupture / 31% death ; bilateral 50%
Femoral	Complications if >3.5cm /thrombus; 88% synchronous aneurysm ; infected IVDU pseudo = most dangerous
Splenic (visceral)	Commonest visceral (~60%); women, portal HTN; pregnancy = rupture
Renal	0.1%; FMD, saccular; 3rd-trimester rupture: 70%/100% mortality
Subclavian	Thoracic outlet repetitive trauma
Pseudoaneurysm Rx	US-guided compression / thrombin injection ; open if infected

Bold = high-yield.